

5633032

B.Tech. DEGREE EXAMINATION, MARCH 2021.

Third Semester

Computer Science Engineering

DATA STRUCTURES

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Define data structures.
2. What is space and time complexity?
3. List the types of linked list.
4. What is polish notation?
5. List four properties of binary tree.
6. Differentiate binary search tree and AVL tree.
7. Define graph and its types.

8. Differentiate Depth first traversal and Breadth first traversal in graphs.
9. Define Hash table.
10. Brief external sorting.

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

All questions carry equal marks.

UNIT I

11. Write the selection sort technique. Illustrate the same with an example that uses 7 different numeric elements.

Or

12. Discuss in detail about algorithmic notation.

UNIT II

13. Explain various operations of queue.

Or

14. List the steps to convert the infix to postfix expression. Illustrate the same using stack behaviour on any example of your own.

UNIT III

15. Discuss the various types of trees. Also, list any 7 terminologies that are associated with binary trees.

Or

16. Explain with proper illustration on insertion of a node in the binary search tree.

UNIT IV

17. Discuss Dijkstra Algorithm for finding shortest path in the graph. Support your explanations with suitable example.

Or

18. Write algorithm that uses Prim's way of deriving minimum cost Spanning tree from a graph.

UNIT V

19. Explain in detail, (a) Reversed tables (b) Symbol tables.

Or

20. Give detailed notes on external storage devices.